

Over the internship I designed and built **PULSE** from the ground up — a complete oil-trading research terminal with a live dashboard. It pulls in **~35 data feeds** (prices, forward curves, inventories, positioning, macro and news), learns what each oil **spread** (the price gap between two futures months) should be worth in the current market, and trades the difference. On top of that sit two more signal engines — one that **predicts how prices react to the weekly US oil-inventory report**, and one that **reads geopolitical news and works out which oil prices it will move** — all shown across a 8-tab dashboard and wired into a **desk that opens and closes paper trades on its own, all day**. Everything below was tested on years of history and reported honestly, net of trading costs.

~35 LIVE DATA FEEDS	6 × 27 SPREADS × MARKET REGIMES	82.9% BACKTEST WIN RATE	37 GEO ASSETS MAPPED	3-way EIA INVENTORY MODEL	284 AUTOMATED TESTS	24/7 DEPLOYED LIVE
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01 THE CORE STRATEGY — A MODEL FOR EVERY MARKET

The heart of PULSE estimates the **fair value** of six oil spreads (Brent & WTI near-month, far-month and butterflies), learns what each oil **spread** (the price gap between two futures months) should be worth in the current market, and trades the difference, but adapted to conditions instead of one-size-fits-all.

- 27 market "regimes."** I split the market on three axes — the **shape of the curve** (oversupply vs shortage), **inventory level** (well-stocked vs tight vs average) and **volatility** (calm/normal/stressed). Three buckets on each = **27 distinct market conditions**, and each spread gets its own model in each one.
- 7 models compete in every cell.** Four linear models (Ridge, Lasso, ElasticNet, Huber) and three gradient-boosted trees (**XGBoost**, **LightGBM**, **CatBoost**) all train on the same data; the one with the best out-of-sample score is picked automatically for that cell — so the method is different where the market behaves differently.
- What actually wins:** across the book **CatBoost** and **Huber** are chosen most often — CatBoost is right **67.5%** of the time where it wins (Sharpe 1.16), Huber elsewhere for stability.
- Each model reads **22–24 point-in-time inputs** (curve shape, trader positioning, refinery margins, macro/rates, volatility) and outputs a fair value **plus a confidence band**, so trade size scales with conviction.

02 BACKTESTED PERFORMANCE (NET OF COSTS)

Tested with an honest expanding-window backtest over **2018–2026** — the model only ever sees past data and is re-fit every quarter (no hindsight):

Win rate — completed trades	82.9%
Held up on a separate test period	~72%
Profit factor (winnings ÷ losses)	≈ 2.0×
Direction called correctly	61.3%
Trades tested · avg hold	6,140 · 3–6 d
Cumulative net P&L (spread pts)	+848

03 RESULTS BROKEN DOWN BY SPREAD

Spread	Win	Sharpe	Net P&L
Brent butterfly	74.7%	1.26	+229
WTI butterfly	71.7%	0.80	+135
Brent M1–M2	62.8%	0.67	+292
WTI far months	58.1%	0.31	+128
WTI M1–M2	57.1%	0.26	+90
Brent far months	52.1%	−0.04	−25

The butterflies are the standouts; I kept the one weak spread (Brent far-months) in the table rather than hide it — honesty over a prettier average.

04 STRATEGY RESEARCH — I TESTED 10 VERSIONS

A careful program to answer: does all this market-condition intelligence actually beat a simple version? Ten variants, all scored the same way:

Version tested	Sharpe
Market-condition model (regime-as-feature)	0.380
Selective version (only where it helps)	0.374
Simple baseline (plain mean-reversion)	0.372
Machine-learned regimes (HMM)	0.289
Volatility-targeted book (GARCH-sized)	0.285

Honest finding: the fancy versions *tied* the simple one rather than beating it — a real result I reported straight. The value was learning *where* complexity helps (WTI front spreads) and using it only there.

05 PREDICTING THE WEEKLY INVENTORY REACTION (EIA)

Each Wednesday the US publishes oil-inventory data that moves the market. I built a model that predicts the reaction for **crude, gasoline and distillate** separately.

- It measures the **surprise** vs the real analyst forecast (10+ years of data), pulls the **actual number live from the official EIA feed**, and previews it with the Tuesday industry estimate.
- It only calls a trade when it has an edge** — a crude surprise is 75–81% predictive in an oversupplied market but a coin-flip when tight, so it stays quiet and **points to gasoline instead**. I also proved US inventories move **WTI ~17× more than Brent**.

11 HOW THIS FITS INTO A REAL AUTOMATED OIL-TRADING DESK

PULSE is already built the way an **automated (algorithmic) trading desk** works — the paper engine becomes a live execution engine simply by swapping the paper broker for a real one. In production it would:

- Generate signals non-stop** from live prices, the inventory calendar and the news wire — the three engines feed one ranked list of trades.
- Auto-execute with built-in risk controls** — volatility-based sizing, a correlation filter against concentration, hard stop-losses, and a shock breaker that halts trading in a crisis.
- Trade the whole complex** — calendar spreads, butterflies, refinery margins (cracks) and grade spreads (WTI–Brent), exactly the products an oil desk runs.
- Keep a human in the loop** — every automated decision comes with its reasoning, confidence and track record, so a trader can supervise or override it.

06 TURNING NEWS INTO TRADES — AI + RETRIEVAL (FLAGSHIP)

The piece I'm proudest of, and something I hadn't seen done before: a system that reads a geopolitical **headline**, works out which **physical oil asset** it's about, and predicts which specific **oil prices** will move, and in which direction.

- A map of 37 real assets** — shipping chokepoints (Hormuz, Suez), refineries, pipelines and producers, each with its location and how a disruption there moves prices.
- AI reads the headline.** A large language model extracts *what* happened, *where* and *how serious* — and correctly separates "Hormuz closes" from "Hormuz reopens" (opposite trades). A keyword reader backs it up so it never breaks without an AI key.
- "This looks like..." — retrieval (RAG).** For any new headline it finds the most similar past events and shows what each actually did to prices — a transparent, explainable approach, not a black box.
- A plain-English desk note.** The AI writes a 2–3 sentence read grounded strictly in the real retrieved numbers — it is never allowed to invent a figure.
- A live world map** lights up each asset as related news breaks.

07 WHAT THE NEWS ACTUALLY MOVED (MEASURED)

Graded on the 2026 Iran/Hormuz crisis (112 events, 1,400 price observations) — these reactions were real and statistically significant:

Measured effect	accuracy	signif.
Chokepoint scare → Brent gains vs WTI	59%	p=0.008
Chokepoint scare → diesel margin firms	57%	p=0.047
Crude price spike fades within a week	reverts	p<0.001

A tradeable pattern emerged: **structural spreads keep the move while flat-price panic fades**. The model only prints a firm number when the statistics support it; otherwise it labels it an estimate.

08 RISK CONTROLS & THE LIVE TRADING ENGINE

- Never doubles up on the same bet** — a correlation filter blocks two trades that are really one risk, and keeps trades that hedge each other.
- Sizes down when markets get wild** — a volatility model (GARCH) shrinks positions as risk rises, cutting the worst drawdown by ~60% and roughly tripling that book's risk-adjusted return.
- A shock detector** spots market stress and pauses new trades automatically.
- Runs live** — reads the desk's 15-minute price feed, re-prices on the current market, logs every signal with its reasoning and confidence, and opens/closes paper trades on its own during market hours.

09 ORIGINAL IDEAS I CONTRIBUTED

- The news-to-price engine itself** — connecting a place in the news to the exact tradeable spread it affects is a genuinely new framework.
- Explainable retrieval (RAG) without a black box** — matching events by their real features, so every recommendation can be audited.
- "Only speak when you have an edge"** — used across all three engines: a firm signal when history backs it, otherwise clearly "no edge here."
- Bridging stale daily data to the live market** so the models score on today's prices in real time.

10 HOW CAREFULLY IT WAS TESTED

- No hindsight, anywhere** — models only ever see past data; results are out-of-sample and after realistic trading costs.
- Everything is checked** — the model's confidence lines up with reality (higher-conviction trades really do work more often), guarded by **284 automated tests**.
- Honesty over polish** — every number traces to a named source, stale data is flagged, and results that didn't work are reported plainly.

12 KEY LESSONS & THE SKILLS I BUILT

- **A simple, well-built strategy is hard to beat** — added complexity only pays in specific, identifiable spots.
- **Fundamentals & geopolitics only matter in the right conditions** — inventories in a glut, news in structural spreads.
- **Knowing when NOT to trade is itself an edge** — precision beats volume.

Skills:	machine learning for time series	backtesting & walk-forward validation	volatility modelling (GARCH)
	market-regime detection	portfolio & risk construction	statistics / hypothesis testing
	large language models	retrieval (RAG)	Python · sklearn · XGBoost/LightGBM/CatBoost · DuckDB
	full-stack (Flask + React)	live deployment	